Software Requirements Specification

PUBLIC TRANSPORT TICKETING

Structured technical document aligned for formal review, sign-off, and handover.

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Req2Design IEEE 830-1998 aligned Generated draft for expert review

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Revision History

Name	Date	Reason For Changes	Version
Abdul Haq	2026-04-18	Initial generated draft for review.	1.0

1: Introduction

1.1: Purpose

Purpose: To define the software requirements for a public transport ticketing system that supports passenger self-service, conductor validation, and operational monitoring functionalities.

1.2: Scope

Scope: This Software Requirements Specification (SRS) covers the core functionalities required by passengers, conductors, and operations staff within the context of a public transport network.

1.3: Definitions/Acronyms

- **PTT:** Public Transport Ticketing
- **QR:** Quick Response code
- **NFC:** Near Field Communication

1.4: References

References: IEEE Std 830-1998, IEEE Standard for Software Requirements Specifications

1.5: Overview

Overview: This document outlines the detailed requirements for the PTT system, including user interactions, system behaviors, and non-functional requirements.

2: Overall Description

Perspective: The perspective of this specification is to provide a comprehensive set of requirements for the PTT system, ensuring it meets the needs of all stakeholders involved.

Functions: The system will perform the following key functions:

- Route planning
- Ticket purchase (single and monthly)
- Ticket validation using QR/NFC
- Offline validation by conductors
- Fare zone management
- Concession category handling
- Penalty calculation
- Monitoring of route usage and traffic patterns
- Device health tracking
- Refund workflow

Users: Primary users include passengers, conductors, and operations teams. Secondary users may include administrative personnel responsible for setting policies and managing the system.

2.4: Constraints

Constraints: The system must operate reliably under varying environmental conditions and ensure data integrity during synchronization processes.

2.5: Assumptions/Dependencies

Assumptions/Dependencies: Assumptions include stable internet connectivity for most transactions but allowance for offline functionality. Dependencies include integration with existing infrastructure such as payment gateways and station hardware.

3: Specific Requirements

External Interfaces:

- Passenger interface via mobile app/web portal
- Conductor handheld devices
- Station kiosks
- Backend servers for processing and storage

3.2: Functional Requirements

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- Route Planning

Input: User's origin and destination locations.

Processing: Generate optimal routes considering available transportation modes and fare zones.

Output: Displayed routes along with estimated costs and travel times.

Priority: High

- Users can select preferred travel options based on cost, time, and mode of transport.
- Real-time updates reflecting current traffic conditions.

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- Ticket Purchase

Input: Passenger details, selected route, chosen ticket type (single or monthly pass), payment method.

Processing: Validate input, calculate total cost, process transaction, generate unique ticket identifier.

Output: Confirmation message displayed on screen, ticket ID sent via SMS/email.

Priority: High

- Support multiple payment methods including cash, credit cards, and digital wallets.
- Provide clear error messages for invalid inputs.

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- Ticket Validation

Input: Passenger presents ticket via QR/NFC reader.

Processing: Verify ticket validity, check against fare zones and concession categories.

Output: Validated status displayed on reader, access granted or denied.

Priority: Medium

- Ensure seamless validation experience without requiring manual intervention.
- Handle various types of tickets including single rides, day passes, etc.

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- Offline Conductor Validation

Input: Conductor scans ticket barcode.

Processing: Check ticket information locally stored on device.

Output: Validity result shown on device display.

Priority: Medium

- Allow quick validation even when no internet connection is available.
- Sync validation records upon re-establishment of network connection.

Non-Functional Requirements

Usability: The system shall be intuitive and easy to navigate for both passengers and operators.

Reliability: The system must function correctly despite potential disruptions in network connectivity.

Performance: The system shall respond promptly to user actions and complete tasks efficiently.

Portability: The system components must be deployable across different environments and devices.

4: System Features

- Mobile application for passenger self-service
- Web portal for advanced queries and account management
- Handheld devices for conductors to scan tickets and log activities
- Kiosk terminals at stations for purchasing and validating tickets
- Centralized backend server for data storage and analytics
- Integration with third-party payment systems
- Data synchronization mechanisms between offline and online states

End of Document.